

QUESTION NO#01:

```
#include <iostream>
```

```
using namespace std;
```

```
class ComplexNumber {
```

```
private:
```

```
    double* real;
```

```
    double* imag;
```

```
public:
```

```
    ComplexNumber(double r, double i) {
```

```
        real = new double(r);
```

```
        imag = new double(i);
```

```
    }
```

```
    ComplexNumber(const ComplexNumber& source) {
```

```
        real = new double(*source.real);
```

```
        imag = new double(*source.imag);
```

```
    }
```

```
    ~ComplexNumber() {
```

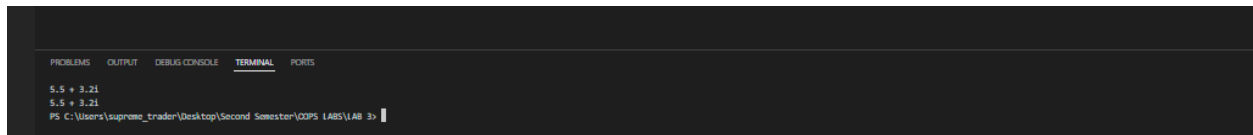
```
        delete real;
```

```
        delete imag;
```

```
    }
```

```
void display() const {  
    cout << *real << " + " << *imag << "i" << endl;  
}  
};
```

```
int main() {  
    ComplexNumber num1(5.5, 3.2);  
    ComplexNumber num2 = num1;  
    num1.display();  
    num2.display();  
    return 0;  
}
```



QUESTION NO#02:

```
#include <iostream>
```

```
using namespace std;
```

```
class Character {
```

```
private:
```

```
    int *health, *attackPower, *defense;
```

```
public:
```

```
    Character() {
```

```
    health = new int(100);  
    attackPower = new int(10);  
    defense = new int(5);  
}
```

```
Character(int h, int a, int d) {  
    health = new int(h);  
    attackPower = new int(a);  
    defense = new int(d);  
}
```

```
Character(const Character& source) {  
    health = new int(*source.health);  
    attackPower = new int(*source.attackPower);  
    defense = new int(*source.defense);  
}
```

```
~Character() {  
    delete health;  
    delete attackPower;  
    delete defense;  
}
```

```
void setHealth(int h) { *health = h; }  
void setAttackPower(int a) { *attackPower = a; }  
void setDefense(int d) { *defense = d; }
```

```
int getHealth() const { return *health; }

int getAttackPower() const { return *attackPower; }

int getDefense() const { return *defense; }


void displayStats() const {

    cout << "H: " << *health << " A: " << *attackPower << " D: " << *defense << endl;

}

};


int main() {

    Character p1;

    Character p2(200, 30, 20);

    Character p3 = p2;

    p3.setHealth(150);

    p1.displayStats();

    p2.displayStats();

    p3.displayStats();

    return 0;

}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
H: 100 A: 10 D: 5
H: 200 A: 30 D: 20
H: 150 A: 30 D: 20
PS C:\Users\supreme_trader\Desktop\Second Semester\OOPS LABS\LAB 3> |
```

QUESTION NO#03:

```
#include <iostream>

#include <iomanip>

using namespace std;

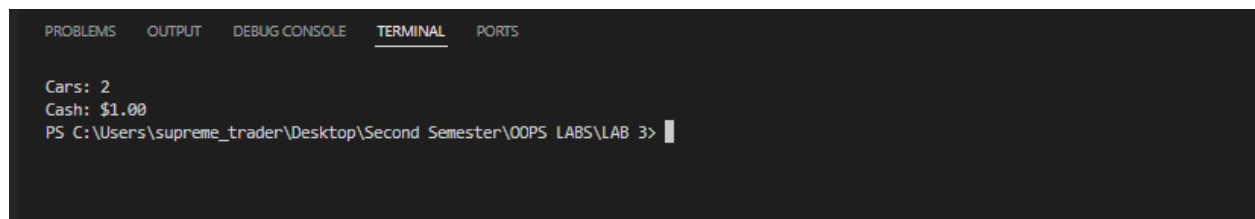
class Tollbooth {
private:
    int totalCars;
    double totalCash;

public:
    Tollbooth() : totalCars(0), totalCash(0.0) {}

    void payingCar() {
        totalCars++;
        totalCash += 0.50;
    }

    void display() const {
        cout << "Cars: " << totalCars << endl;
        cout << "Cash: $" << fixed << setprecision(2) << totalCash << endl;
    }
};
```

```
int main() {  
    Tollbooth booth;  
    booth.payingCar();  
    booth.payingCar();  
    booth.display();  
    return 0;  
}
```

A screenshot of a C++ IDE's terminal window. The terminal has tabs for PROBLEMS, OUTPUT, DEBUG CONSOLE, TERMINAL (which is active), and PORTS. The output text in the terminal is: "Cars: 2", "Cash: \$1.00", and "PS C:\Users\supreme_trader\Desktop\Second Semester\OOPS LABS\LAB 3>".

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  
  
Cars: 2  
Cash: $1.00  
PS C:\Users\supreme_trader\Desktop\Second Semester\OOPS LABS\LAB 3>
```

QUESTION NO#04:

```
#include <iostream>  
  
#include <string>  
  
using namespace std;  
  
class bookType {  
private:  
    string title;  
    string authors[4];  
    int numAuthors;  
    string publisher;  
    string ISBN;  
    double price;  
    int stock;
```

public:

// Default constructor

bookType() {

 title = "";

 publisher = "";

 ISBN = "";

 price = 0.0;

 stock = 0;

 numAuthors = 0;

}

// Setters

void setTitle(string t) { title = t; }

void setPublisher(string p) { publisher = p; }

void setISBN(string i) { ISBN = i; }

void setPrice(double p) { price = p; }

void setStock(int s) { stock = s; }

void setAuthors(string auth[], int count) {

 numAuthors = (count > 4) ? 4 : count;

 for (int i = 0; i < numAuthors; i++)

 authors[i] = auth[i];

}

// Getters

```
string getTitle() const { return title; }  
string getISBN() const { return ISBN; }  
int getStock() const { return stock; }
```

```
// Operations
```

```
bool isTitleSame(string t) const {  
    return title == t;  
}
```

```
bool isISBNSame(string i) const {  
    return ISBN == i;  
}
```

```
void updateStock(int amount) {  
    stock += amount;  
}
```

```
void showAuthors() const {  
    cout << "Authors: ";  
    for (int i = 0; i < numAuthors; i++)  
        cout << authors[i] << " ";  
    cout << endl;  
}
```

```
void showBookInfo() const {  
    cout << "Title: " << title << endl;
```



```
        cout << "ISBN: " << ISBN << endl;

        cout << "Publisher: " << publisher << endl;

        cout << "Price: " << price << endl;

        cout << "Stock: " << stock << endl;

        showAuthors();

        cout << "-----" << endl;
    }

    // Destructor

    ~bookType() {

        // No dynamic memory used

    }

};

int main() {

    bookType library[100];

    int count = 0;

    string auth1[] = {"Author A", "Author B"};

    library[0].setTitle("C++ Programming");

    library[0].setISBN("123-ABC");

    library[0].setPublisher("TechPress");

    library[0].setPrice(50.0);

    library[0].setStock(10);

    library[0].setAuthors(auth1, 2);

    count++;
```

```
// Search by ISBN

string searchISBN = "123-ABC";

for (int i = 0; i < count; i++) {

    if (library[i].isISBNSame(searchISBN)) {

        library[i].updateStock(5);

        library[i].showBookInfo();

    }

}

// Search by Title

string searchTitle = "C++ Programming";

for (int i = 0; i < count; i++) {

    if (library[i].isTitleSame(searchTitle)) {

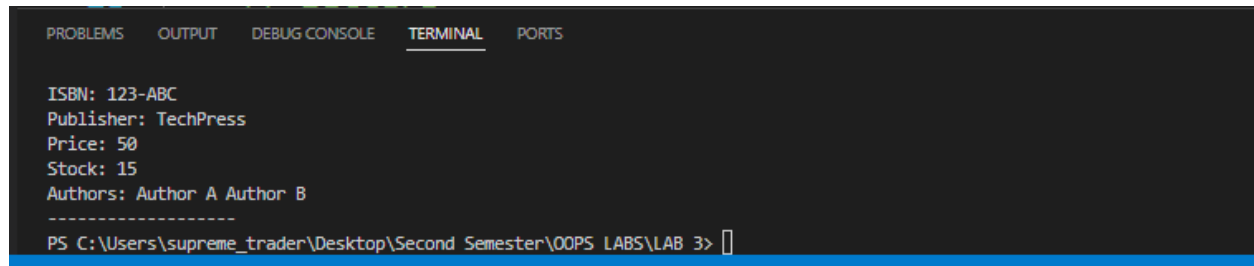
        library[i].showBookInfo();

    }

}

return 0;

}
```



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

ISBN: 123-ABC
Publisher: TechPress
Price: 50
Stock: 15
Authors: Author A Author B
-----
PS C:\Users\supreme_trader\Desktop\Second Semester\OOPS LABS\LAB 3> |
```